

Random

..... Spring Boot + React AI
 Java

.....30.....

"..... Random..... Spring Boot 3.2..... JWT..... MiMo..... AI Vercel + .. Render + PostgreS

■ ■ ■ ■ ■ ■ ■

```

.....
      ..
    React 18 + TypeScript + Vite 6 + Tailwind CSS v4
    React Router v7 + Framer Motion + Axios + Zod
      .
    HTTPS / REST API
.....
      ..
    Spring Boot 3.2.5 + Spring Security + Spring Data JPA
    JWT (JJWT 0.12.5) + BCrypt + Bean Validation
    LLM Integration (OpenAI-compatible API)
.....
      ...
    MySQL 8.0 (..) / PostgreSQL 16 (..)
    Hibernate 6 + ddl-auto=update
.....
      ..
    Docker ..... + Render (..) + Vercel (..)
.....

```

■ ■ ■ ■ ■ ■ ■

3.1 ER ...

```

users .....
.
. 1:N      . 1:N
.
options ... decisions ..1:1... feedback
.
. N:1                      . N:1
.
categories                  .
.
. 1:N                      .
.
user_preferences .....

```

3.2

```
| .. | .. | +-----+-----+ | users | id, username, password_hash, nickname, email | BCrypt .....username ..... |
categories | id, name, display_name, icon, sort_order | .. 5 ..... | options | id, user_id(FK), category_id(FK),
name, tags, is_active | ..... (user_id, category_id) | | decisions | id, user_id(FK), category_id(FK), chosen_
option_id(FK), context_json | context_json ..... | JSONFeedback | id, decision_id(FK, UNIQUE), user_id(FK),
rating(1-3) | ..... | user_preferences | id, user_id(FK), category_id(FK), tag, weight(DECIMAL 5,4) | ..... (user_id,
category_id, tag) |
```

3.3

Q:

A:decisionsis_active.....

Q: ... decisions ... context ... JSON ..

A: JSON @JdbcTypeCode(SqlTypes.JSON) ... PostgreSQL ... JSON

Q: user_preferences ...[0.05, 0.95]..

A: "!"-Exploration-Exploitation..... 0.....0.05 5%0.95 ... 100%

.....

4.1

..DecisionServiceImpl.java..

```
..: userId, categoryId, optionIds(..)
..: DecisionVO { chosenOption, alternates, decidedAt }
1. ....
2. ... optionIds.....
3. ... preferenceFilters.....
4. ... N .....
5. ....:
weight = 1.0
for tag in option.tags:
    weight *= (0.5 + userPreferenceWeight[tag])
if option recently chosen:
    weight *= 0.3
weight += random(0, 0.1) // ....
6. ....:
total = sum(weights)
rand = random(0, total)
cumulative = 0
for option in candidates:
    cumulative += option.weight
    if cumulative >= rand:
        return option
7. .... 2 .....
8. ....
```

∴ $\Theta(n \times m) \cdot n \dots m \dots$

....

Q:

A:
.....

Q:

A:0.1 1.0

4.2

..... ..
.....
.. (3) $w += 0.05 \times (1 - w)$.. 1.0
.. (2)
... (1) $w -= 0.05 \times w$.. 0.0

....

- 0.....
- ..[0.05, 0.95].....
-
.....

....

Q: 0.05

A: 0.05 0.2 0.01 0.05 14 0.5 ... 0.85.....

..Spring Boot

5.1

Controller · Service · Repository · Entity
·
·
 DTO/VO
(....) (....)

| .. | .. ||-----||-----|| Controller |@RestController + Bean Validation || Service |@Service +
@Transactional || Repository | | Spring Data JPA + ... || Query | | JPA Entity + Hibernate |

5.2JWT.

....:
Client · Authorization: Bearer <token> · JwtAuthFilter
 .. token .. userId .. SecurityContext
 Controller: auth.getPrincipal()
Token ..:
Header: { "alg": "HS256" }
Payload: { "sub": "123", "username": "test", "iat": ..., "exp": ...

```
}
Signature: HMAC-SHA256(header + payload, secret)
```

....

Q: JWT · Session ······ JWT·

A: Session ······ Session· Redis·JWT ···Token ······ Render ······ JWT ······

Q: JWT ······

A: JWT ······ Token ······1) ······ + Refresh Token ···2) ··· Token ···Redis·3) · Token ······

5.3

```
@RestControllerAdvice
public class GlobalExceptionHandler {
    @ExceptionHandler(BusinessException.class)    // ....
    @ExceptionHandler(MethodArgumentNotValidException.class) // ....
    @ExceptionHandler(Exception.class)           // ....
}
```

··{··code: 0, msg: "...", data: null }··code··

5.4

```
application.yml          # .... (Jackson, JPA)
application-dev.yml      # ..: MySQL, show-sql=true, LLM=Mimo
application-prod.yml     # ..: PostgreSQL, LLM from env vars
```

·spring.profiles.active ······JWT Secret·API Key······

··AI ·····

6.1

```
AiController · LlmService · LlmConfig · HTTP Client · Mimo API
·
... getSmartRecommendation() // ....
... suggestNewOptions()      // .....
... analyzePreferences()     // ....
... explainDecision()        // ....
```

6.2

- **API** ··OpenAI Chat Completions ······ LLM ···
- ···3 ··· + ····2s, 4s, 6s······
- ···60 ·······
- **Prompt** ······ prompt·····

6.3

Q:

A: LLM AI WebSocket SSE

Q: .. LLM API

A: LlmServiceImpl.chat() ..null..Controller "AI"..... LLM.....

.....

7.1 Docker

```
# Stage 1: Build
FROM maven:3.9-eclipse-temurin-17 AS build
COPY random-backend/pom.xml .
RUN mvn dependency:go-offline -B          # .....
COPY random-backend/src ./src
RUN mvn clean package -DskipTests -B
# Stage 2: Runtime
FROM eclipse-temurin:17-jre-alpine      # .....
COPY --from=build /app/target/*.jar app.jar
COPY start.sh /app/start.sh
CMD ["/app/start.sh"]
```

....

Q:

A: Maven JDK 800MB JRE + JAR .. 200MB 75%

7.2

```
# start.sh
exec java -Duser.timezone=Asia/Shanghai
-Dserver.port=${PORT:-10000} -jar app.jar
```

- UTC
- ..\$PORT Render/Railway · PaaS ..
- ../api/health ...Render

7.3

| .. | .. || | | | | Railway .. 502\$PORT · Dockerfile CMD start.sh .. shell | PostgreSQL JSON ... |
Hibernate .. varchar .. json @JdbcTypeCode(SqlTypes.JSON) | | Render | ... 30-60 | Axios AI ... 60s |
Render · UTC 8 .. | Duser.timezone=Asia/Shanghai | | AI | .. HttpClient ... + | HttpClient +
synchronized + 3 ... |

.....

8.1

- `Controller · Service · Repository`
- `{code, msg, data}`
- `@RestControllerAdvice`
- `Profile`

8.2

- `O(n)`
- `[0:05, 0.95]`

8.3

- `BCrypt` + `JWT` + `Spring Security`
- `JSON`
- `Docker`
- **DevOps** `CI/CD` `GitHub` `Render`

8.4

- `JVM`
- `+` `+` `+`
- `MySQL` `PostgreSQL`

8.5 AI

- **LLM** `OpenAI` `Prompt`
- `LLM`
-

.....

9.1 Spring Boot

Q: Spring Boot

A: `@SpringBootApplication` `@EnableAutoConfiguration` `spring.factories`
`Spring Boot` `AutoConfiguration.imports` `@ConditionalOnClass`
`@ConditionalOnMissingBean` `spring-boot-starter-data-jpa`
`Hibernate` `DataSource` `EntityManager`

Q: Spring Security

A: `Spring Security` `FilterSecurityContextPersistenceFilter` ·
`UsernamePasswordAuthenticationFilter` `ExceptionTranslationFilter` ·
`FilterSecurityInterceptor` `UsernamePasswordAuthenticationFilter`
`JwtAuthFilter` `Token` ·

9.2

Q: JPA · MyBatis JPA·

A: JPA · ORM SQL··· CRUDMyBatis · SQL ~~SQL~~ByXxx·@Query····JPA···

Q: Hibernate · N+1

```
A: ...FetchType.LAZY.....@EntityGraph .....2) JOIN FETCH @BatchSize
..... Decision-category-chosenOption ...@Query + JOIN FETCH .....
```

9.3

Q:

```
A: ·synchronized ······@VerdisSETNX·····+ ······READ COMMITTED·····
```

Q:

A: 1) (user_id, category_id) 2) 3) Redis ...

9.4 ...

.....

- `..+lmService .. +lmServiceImpl .....` LLM ...
- `..... Entity@PrePersist / @PreUpdate ..`
- `..... VOBuilder`
- `.....Spring Security · Filter Chain`
- `.....Spring@Bean ..`

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- ```

1 Redis
2 LLM RabbitMQ/Kafka ..
3 Elasticsearch
4 AI
5 Prometheus + Grafana .. API
6 JUnit 5 + Mockito
7 API ..Swagger/OpenAPI
8 WebSocket ..LLM AI

```

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|      |       |        |         |          |      |      |        |      |        |                |                              |
|------|-------|--------|---------|----------|------|------|--------|------|--------|----------------|------------------------------|
| ·    | -     | ---    | -----   | · Java · | 56 · | 44 · | 6API · | 20 · | 4000 · | MiMo LLM API · | Vercel + Render + PostgreSQL |
| ···· | ····· | ······ | ······· |          |      |      |        |      |        |                |                              |

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